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# Mediacrest Training College

Nairobi, Kenya

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## Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

### Comprehensive Session Notes

#### Week 4 · Day 1

*The Clinical Decision-Making and Policy Focus: Logistic Regression for Heart Disease Prediction*

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**Session Date:** 1st September 2026

**Mode:** Hybrid – Online / Physical (EAT)

*“A model is only as valuable as the clinical decisions it informs. Thresholds must be chosen based on clinical context, not software defaults.”*

## About This Document

These comprehensive notes accompany **Week 3, Day 2** of the **Advanced Data Analytics & Machine Learning** program at Mediacrest Training College. Building on the statistical rigor and diagnostics established in Day 1, this session pivots to **Clinical Decision-Making and Policy**.

We will explore how to translate complex statistical outputs into actionable insights for non-technical stakeholders. Key topics include clinical Exploratory Data Analysis (EDA), stratified prevalence profiling, building parsimonious risk models, avoiding the "Odds Ratio vs. Risk Ratio" trap, and scenario-based threshold selection (e.g., Primary Care screening vs. Specialist referral). The session concludes with actionable recommendations for integrating predictive models into national Health Management Information Systems (HMIS).

## Course at a Glance

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
|            | 8 Weeks (20+ intensive sessions)                                                       |
| Duration   |                                                                                        |
| Core Tools | Python (pandas, NumPy, statsmodels, scikit-learn, matplotlib, seaborn)                 |
| Focus      | Clinical EDA, parsimonious modeling, threshold optimization, health policy integration |
| Capstone   | Production-ready ML system with diagnostics, explainability & policy recommendations   |
| Context    | Healthcare, Clinical Risk Prediction, and Epidemiological Modeling                     |

## Where This Session Fits

Week 3 is titled “**Statistical Rigor, Diagnostics & Advanced Modeling.**”

- Day 1:** Advanced Data Prep, Logistic Regression, Odds Ratios, Diagnostics (VIF, Cook’s D, Linearity of Logits, Interactions), and Evaluation Metrics.
- Day 2 (this document):** Clinical EDA, Risk Factor Profiling, Parsimonious Modeling, Threshold Selection for Clinical Policy, and HMIS Integration.
- Day 3:** Advanced Classification (Random Forests, Gradient Boosting) and model interpretability (SHAP values).

## How to Use These Notes

- Follow along in code.** Every listing is runnable. Open the companion Jupyter Notebook and execute each cell as you read.
- Boxes matter.** **Key Concept** boxes hold the theory; **Warning** boxes highlight common pitfalls; **Kenyan Clinical Context** boxes connect material to local healthcare realities.

- **Do the exercises.** Each chapter ends with practice tasks designed to test your statistical intuition and policy reasoning.

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# 1 Clinical EDA & Risk Factor Profiling

## 1.1 Target Audience, Dataset & Dependencies

This module is designed for analysts transitioning to rigorous statistical modeling and clinical decision support. We will use the `Heart.csv` dataset, containing clinical cardiovascular data. The required Python dependencies are: `pandas`, `numpy`, `statsmodels`, `scikit-learn`, `matplotlib`, and `seaborn`.

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 import statsmodels.api as sm
6 import statsmodels.formula.api as smf
7 from sklearn.model_selection import train_test_split
8 from sklearn.metrics import precision_recall_curve, auc, classification_report,
   confusion_matrix
9
10 # Set professional plotting style
11 sns.set_theme(style="whitegrid", context="talk")
12 plt.rcParams['figure.figsize'] = (10, 6)
13 plt.rcParams['font.family'] = 'sans-serif'
14
15 # 1. Load Data, explicitly handling 'NA', '?', 'NaN', and '' as missing values
16 df = pd.read_csv('Heart.csv', na_values=['NA', '?', 'NaN', ''])
```

Listing 1: Loading the Needed Libraries and Setting Professional Style

## 1.2 Handling Missingness: Clinical Reality

### Understanding Missing Data Mechanisms

- **MCAR (Missing Completely at Random):** Missingness is unrelated to any observed or unobserved data.
- **MAR (Missing at Random):** Missingness is related to observed data (e.g., older patients might systematically skip the `Ca` test).
- **MNAR (Missing Not at Random):** Missingness is related to the unobserved value itself (e.g., patients with severe disease avoid the `Thal` scan).

### Clinical Reality

In clinical data, MAR or MNAR is common. We must impute cautiously to avoid introducing bias or losing valuable "missingness" information.

## 1.3 Imputation Strategy & Feature Engineering

An initial audit reveals missing values in `Ca` (4 missing) and `Thal` (2 missing).

- **For `Ca` (Ordinal 0-3):** We use median imputation, which is robust to outliers and preserves the ordinal nature of major vessel count.

- **For Thal (Nominal):** We create an explicit **unknown** category. This preserves the information that the test was missing, avoiding bias.

```
1 # Imputation
2 df['Ca'] = pd.to_numeric(df['Ca'], errors='coerce')
3 df['Ca'] = df['Ca'].fillna(df['Ca'].median())
4 df['Thal'] = df['Thal'].fillna('unknown')
5
6 # Feature Engineering: Age Bins for stratified analysis
7 df['Age_Bin'] = pd.cut(df['Age'],
8     bins=[0, 50, 60, 70, 100],
9     labels=['<50', '50-60', '61-70', '>70'])
10
11 # Feature Engineering: Sex Labels for intuitive reporting
12 df['Sex_Label'] = df['Sex'].map({0: 'Female', 1: 'Male'})
```

Listing 2: Imputation and Feature Engineering

### Pro Tip

Avoid mean imputation for ordinal clinical scores. Preserving "missingness" as a distinct category in nominal variables prevents the model from making unfounded assumptions.

## 1.4 Stratified Prevalence Analysis

Before modeling, we must understand the baseline prevalence of the target variable (HD). This cohort is relatively balanced (approx. 50% prevalence).

- **By Chest Pain:** Patients presenting with **asymptomatic** chest pain often show the highest prevalence of underlying heart disease, highlighting the danger of relying solely on patient-reported symptoms.
- **By Age Bin and Sex:** The prevalence of heart disease steadily increases with age across both sexes. However, the rate of increase and baseline risk often diverge, necessitating sex-specific risk assessment protocols in older age brackets (>70).

### Kenyan Clinical Context

Understanding the baseline sex and age distribution is the first step in ensuring our final model does not perpetuate or amplify algorithmic bias against underrepresented groups during deployment in diverse Kenyan demographics.

## 2 Logistic Regression for Risk Stratification

### 2.1 Defining the Parsimonious Clinical Risk Model

A **parsimonious** model includes only the most predictive and clinically relevant variables, avoiding overfitting. We utilize the `statsmodels` formula API for clear, R-style statistical specification.

```
1 # Drop missing values for this specific model fit demonstration
2 df_clean = df.dropna()
3
4 formula = "HD ~ Age + Sex + C(ChestPain) + RestBP + Chol + Fbs + C(RestECG) +
           MaxHR + ExAng + Oldpeak + C(Slope) + Ca + C(Thal)"
```

Listing 3: Model Specification

### 2.2 Fitting the Logistic Regression Model

We fit the model using Maximum Likelihood Estimation (MLE). Setting `disp=0` suppresses verbose convergence output for cleaner reporting.

```
1 model = smf.logit(formula, data=df_clean).fit(disp=0)
2 print(model.summary())
```

Listing 4: Fitting the Model via MLE

#### Key Output Interpretation:

- **No. Observations:** 297 (after dropping NAs).
- **Pseudo R-squared (McFadden):** 0.5325. This indicates a strong fit, showing substantial improvement over an intercept-only (null) model.
- **LLR p-value:**  $2.124 \times 10^{-36}$ . The model fits the data significantly better than the null model.

### 2.3 Translating Model Outputs for Stakeholders

Statistical outputs (log-odds, p-values) are not intuitive for clinicians, patients, or policymakers.

#### Translation Strategies

1. **Odds Ratios (OR):** Useful for researchers to understand relative effect sizes.
2. **Marginal Predicted Probabilities:** The gold standard for patient-facing risk communication (e.g., "You have a 25% estimated risk").
3. **Absolute Risk Differences:** Clearly shows the impact of an intervention or risk factor.

## 2.4 The Prevalence Trap: OR vs. RR

### The Prevalence Trap

- **Odds Ratio (OR):** Native to Logistic Regression. Compares odds.
- **Risk Ratio (RR):** Compares probabilities (Risk). More intuitive for humans.

If the outcome is rare ( $< 10\%$ ),  $OR \approx RR$ . However, Heart Disease in this dataset is  **$\sim 50\%$  prevalent**. Therefore, the OR will **grossly exaggerate** the effect size compared to the RR.

### Kenyan Clinical Context

In clinical reporting (e.g., at Kenyatta National Hospital or MP Shah), do not tell a patient they are "8 times more likely" to have heart disease based on an OR. Explicitly state that OR is not RR, and use marginal predicted probabilities for patient-facing communication.



## 3 Model Evaluation & Clinical Trade-offs

### 3.1 Train/Test Split Strategy

To avoid overly optimistic in-sample metrics and assess generalizability to unseen patients, we split the data.

```
1 # Note: In production, X should exclude engineered columns like 'Age_Bin' if not
   in the model
2 X = df_clean.drop(columns=['HD', 'Age_Bin', 'Sex_Label'])
3 y = df_clean['HD']
4
5 X_train, X_test, y_train, y_test = train_test_split(
6     X, y, test_size=0.25, random_state=42, stratify=y
7 )
```

Listing 5: Train/Test Split

#### Note on Demonstration

For the simplicity of visualizing curve shapes in this script, full dataset probabilities are sometimes used. In strict production, predictions must be mapped exclusively through `X_test`.

### 3.2 ROC-AUC and Precision-Recall Curves

- **ROC-AUC (0.935):** Plots True Positive Rate (Sensitivity) vs. False Positive Rate (1 - Specificity). An AUC of 0.935 indicates an excellent ability to distinguish between patients with and without heart disease.
- **Precision-Recall AUC (0.929):** Crucial for high-stakes clinical data where False Positives (unnecessary angiograms) and False Negatives (missed diagnoses) carry heavy, asymmetric costs. It demonstrates strong predictive performance in balancing Precision (Positive Predictive Value) and Recall (Sensitivity).

#### Clinical Relevance

In heart disease screening, a high Precision-Recall AUC ensures that when the model flags a patient as high-risk, it is highly likely to be a true positive, minimizing unnecessary, invasive follow-up procedures at resource-constrained facilities.

## 4 Threshold Selection for Clinical Policy

### 4.1 The Challenge of Default Thresholds

Machine learning libraries default to a classification threshold of **0.5**.

- **The Problem:** A 0.5 threshold assumes that the cost of a False Positive (unnecessary test) is exactly equal to the cost of a False Negative (missed diagnosis).
- **Clinical Reality:** These costs are rarely equal. The optimal threshold depends entirely on the **clinical context** and **resource availability**.

#### Policy Directive

Thresholds must be explicitly chosen based on the specific healthcare scenario, not left to software defaults.

### 4.2 Scenario A: National Screening Program (Primary Care)

#### The Clinical Goal

**Maximize Recall (Sensitivity).** We want to catch as many true cases of heart disease as possible in the general population.

- **Trade-off:** We accept more False Positives.
- **Rationale:** In primary care, it is cheaper and safer to re-test a healthy patient (False Positive) than to miss a life-threatening condition (False Negative).
- **Threshold Strategy:** Shift the threshold **lower** (e.g.,  $< 0.5$ ) to cast a wider net, ensuring high sensitivity. Ideal for Community Health Promoters (CHPs) or mobile screening clinics.

### 4.3 Scenario B: Specialist Referral (Cardiology Clinic)

#### The Clinical Goal

**Maximize Precision.** We want to ensure that patients referred for invasive, expensive procedures (like angiograms) truly have the disease.

- **Trade-off:** We accept more False Negatives.
- **Rationale:** In a specialist setting, the cost (financial and physical) of an unnecessary invasive procedure is very high. We must be highly confident before acting.
- **Threshold Strategy:** Shift the threshold **higher** (e.g.,  $> 0.5$ ) to ensure only the highest-risk patients are flagged.

```
1 # Example: Finding threshold for ~85% precision
2 # (Conceptual code for threshold optimization)
3 target_precision = 0.85
4 # ... optimization logic yields:
5 # Recommended Probability Threshold: 0.658
```

```
6 # Expected Precision: 85.3%
7 # Expected Recall (Sensitivity): 82.9%
```

Listing 6: Targeting High Precision

#### 4.4 Cost-Benefit Analysis of Thresholds

Threshold selection is fundamentally an economic and ethical optimization problem.

- **Example Cost:** A False Positive Angiogram may cost approximately **20,000 Ksh**, plus patient anxiety and procedural risk.
- **Example Cost:** A False Negative may result in an unmanaged myocardial infarction, with catastrophic health and financial outcomes.

##### Policy Directive

Healthcare administrators must explicitly define the acceptable cost ratio of False Positives to False Negatives. The model threshold should then be mathematically optimized to minimize the *expected total cost*.

## 5 Best Practices, Checklist & Deliverables

### 5.1 Clinical Decision Support Report

The final output of this pipeline is a structured, reproducible report for stakeholders.

CLINICAL DECISION SUPPORT REPORT: Heart Disease Prediction

1. DATA PREPARATION

- Missingness handled: `Ca` imputed with median (0.0); `Tha1` imputed as `unknown`.

- Encoding: Dummy variables applied to nominal features; ordinal features retained.

2. MODEL FIT & ODDS RATIOS

- Pseudo R-squared (McFadden): 0.533

- Top Predictors: Asymptomatic Chest Pain (OR=8.06), Sex (OR=4.69), Reversible Thal (OR=4.03).

3. DIAGNOSTICS

- Multicollinearity: Max VIF = 1.35 (Age). No severe multicollinearity detected.

- Linearity of Logit: Visually inspected and holds.

- Interaction Check: `ExAng * Sex` p-value = 0.238 (Not significant; no effect modification).

4. EVALUATION METRICS

- ROC-AUC: 0.935 | Precision-Recall AUC: 0.929

- Recommended Policy Thresholds: 0.481 (Balanced), 0.658 (High Precision for Specialist Referral).

NOTE ON CLINICAL REPORTING:

Odds Ratios (OR) are reported above. Because the prevalence of Heart Disease in this cohort is high (~50%), the OR will **overstate** the Risk Ratio (RR). Use marginal predicted probabilities for patient-facing risk communication.

### 5.2 Next Steps & Recommendations

Actionable Directives for Health Systems

1. **HMIS Integration:** Integrate this parsimonious model into the national Health Management Information System (HMIS) or develop a lightweight mobile tool for Community Health Promoters (CHPs).

2. **Prospective Pilot:** Conduct a pilot study in 2-3 sub-county hospitals to validate these scenario-specific thresholds on local Kenyan demographic data before nationwide rollout.

3. **Bias Monitoring:** Ensure continuous monitoring for algorithmic bias across gender and age groups. Disparate impact must be actively measured and mitigated.

4. **Model Retraining:** Establish a pipeline for periodic model retraining as new clinical data becomes available, ensuring the model adapts to shifting population health trends.

Final Deliverable: The Policy Memo

Draft a 1-page technical memo addressed to the County Director of Health. Your memo must include:

1. **Model Justification:** Briefly explain why a parsimonious logistic regression was chosen and how missingness was handled to preserve clinical integrity.

2. **Diagnostic Validation:** Report the VIF and interaction check findings, explicitly stating whether the model is stable across demographic groups.
3. **Policy Recommendation:** Recommend specific classification thresholds for (a) Community Health Promoter screening and (b) County Referral Hospital triage, justifying the trade-offs in terms of cost and patient safety.

## 6 Glossary of Key Terms

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**Health Management Information System (HMIS)** A systematic framework for collecting, processing, and using health data to support decision-making at all levels of the health system.

**Maximum Likelihood Estimation (MLE)** The method used to estimate the parameters of a logistic regression model by maximizing the likelihood of observing the given data.

**Odds Ratio (OR)** The ratio of the odds of an event occurring in one group to the odds of it occurring in another group. Prone to exaggeration when outcome prevalence is high.

**Parsimonious Model** A statistical model that achieves a good fit with the fewest possible predictor variables, reducing the risk of overfitting and improving interpretability.

**Precision-Recall AUC** The area under the Precision-Recall curve, highly informative for evaluating models where false positives/negatives have asymmetric costs.

**Risk Ratio (RR)** The ratio of the probability of an outcome in an exposed group to the probability in an unexposed group. Preferred over OR for clinical communication when prevalence is high.

## 7 References & Further Reading

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